

RAHIL MUKHI

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PROFESSIONAL SUMMARY

Master of Information Technology (Software Engineering) student at Charles Darwin University with industry experience across mobile, web and machine learning. Built and released two privacy-focused iOS and Android apps, one now live on the App Store and developed Python ML models for predictive analytics. A collaborative team member and clear communicator, keen to contribute to technology teams in the Northern Territory.

EDUCATION

Master of Information Technology (Software Engineering)

Charles Darwin University

Nov 2024 – Present

GPA: 5.42/7

Bachelor of Computer Engineering

Gujarat Technological University, India

2020 – 2024

CGPA: 8.22/10

TECHNICAL SKILLS

Languages	Python, Swift, Kotlin, Dart, Java, C, JavaScript, SQL, HTML, CSS
Mobile & web	SwiftUI, Jetpack Compose, Flutter, Supabase, PostgreSQL, Claude API
Data & ML	scikit-learn, pandas, NumPy, Apple Vision, ML Kit, Power BI, Tableau, Advanced Excel
Practices	Agile / Scrum, Waterfall, OOP, manual and automated testing, App Store and Google Play releases
Tools	Jira, Confluence, Trello, Figma, Visio, Lucidchart, draw.io, Miro, Microsoft Teams, SharePoint

WORK EXPERIENCE

Motivolog

Jul 2026 – Present

Computer Science Intern

- Built **Task Loop**, an offline-first app for reusable checklists with time- and location-based reminders, as native iOS (SwiftUI) and Android (Kotlin, Jetpack Compose) apps sharing one domain model; now live on the App Store.
- Used OS-level geofencing instead of continuous location polling to keep battery drain negligible, and migrated the app to fully on-device storage, removing accounts and all user-data collection.
- Developed **OneShot**, an on-device duplicate photo and video cleaner using perceptual hashing, colour histograms and ML (Apple Vision, MediaPipe, ML Kit) to group near-identical shots and suggest which to keep.
- Validated detection against a ground-truth photo set, replacing locality-sensitive hashing that missed 30–43% of true duplicate pairs with a parallelised exhaustive comparison, and added star clustering to prevent false groupings.
- Took both apps through App Store and Google Play review, including privacy manifests and data-safety forms.
- Compared open-core licensing and pricing across 11 competing tools to recommend a strategy for is.team.

Spider Communication

Jul 2024 – Oct 2024

Software Trainee

- Contributed to a real estate listing and analytics platform for property listings, enquiries and insights.
- Helped implement Python machine learning models that analyse property trends and power price estimation.
- Supported backend development, data preprocessing, testing and debugging and worked with senior developers on sprint planning, documentation and progress reporting.

ViitorCloud Technologies

Feb 2024 – Apr 2024

Intern Engineer

- Contributed to a predictive analytics model for business insights across data collection, cleaning and training, and built and tested an ML model from scratch under senior guidance.
- Wrote technical documentation and progress reports, delivering modules on schedule in an Agile team.

Amivir Infotech

Jul 2023 – Aug 2023

Web Development Intern

- Helped develop a hotel search and booking website, building responsive HTML, CSS and JavaScript interfaces connected to the booking backend and data storage.
- Tested and debugged the site to ensure smooth navigation, reliable functionality and optimised performance.

PROJECTS

Smart Learn

AI-powered learning platform

- Built a full-stack gamified learning platform for grades 1–10 that uses the Claude API to turn any topic a student enters into a personalised course, with lessons and quizzes generated as schema-validated structured output.
- Designed six interactive quiz types, including drag-to-reorder, match-the-pairs and sort-into-groups, with KaTeX equation rendering and server-side grading that drives XP, levels, streaks, badges and a leaderboard.
- Added an AI feedback loop that reviews a student's answers and generates a revised course targeting weak spots, with Supabase sign-in (email and Google) and Postgres row-level security keeping each student's data private.

Green Miles

Sustainable travel tracking app

- Developed a Flutter mobile app that records walking, cycling, bus and motorbike trips by GPS and calculates carbon savings against a petrol-car baseline using published per-kilometre emission figures.
- Built live route mapping with background tracking, weekly insights, an opt-in community leaderboard, offline-first trip sync, and speed-based checks that validate the selected transport mode.
- Used an MVVM architecture on a Supabase backend (row-level security, triggers, RPCs).

House Price Prediction

Predictive analytics web application

- Developed a predictive model on housing data to estimate property prices, covering data cleaning, feature selection and regression modelling.
- Implemented a web interface that generates real-time price predictions from user inputs.

Brain Tumour Detection

Machine learning web application

- Trained a machine learning model on MRI scan data to detect brain tumours, using data preprocessing and model evaluation to improve prediction accuracy.
- Integrated the model into a web interface where users upload an MRI image and receive an instant result.

LEADERSHIP & COMMUNITY

Student Representative

Your Voice Student Network, Charles Darwin University

- As part of this network, we share our thoughts, ideas and experiences in a way that positively impacts university life. As a representative, I communicate what other students are experiencing, the improvements they need and what is working well.

Committee Member

Better Together Committee, Charles Darwin University

- As part of this program, we are creating a respectful, safe and inclusive campus culture at CDU. I collaborate on student-focused initiatives, awareness activities and events that foster a supportive and inclusive environment.

REFEREES

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